



Gesare-n / EV-charging-optimization--ML

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About



No description or website provided.

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Contributors 1



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Languages

● Jupyter Notebook 100.0%

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Code

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Gesare-n Fix dataset link in README

c9c307c · 9 minutes ago ⌚

.gitignore	readme	3 hours ago
EV_Charging_Deck_Improved.pdf	pitch deck	31 minutes ago
EV_Charging_Optimization_Final (1)....	improved model version	2 hours ago
EV_Charging_Optimization_Merged...	improved model version	2 hours ago
README.md	Fix dataset link in README	9 minutes ago
ev_charging_data.csv	Add files via upload	10 hours ago

📖 README



ML-Driven EV Charging Optimization for Kenya's Power Grid

A machine learning classification model that predicts EV charging priority — **Low / Medium / High** — enabling Kenya Power (KPLC) to manage grid stability, defer low-priority loads to off-peak windows, and reduce peak demand costs. Built as a proof-of-concept for integration with Geviton's SafiVolt battery management system.

Stakeholders: Kenya Power & Lighting Company (KPLC) · Geviton · Enzi · BasiGo

Final Model: Calibrated Logistic Regression (Isotonic) · **Accuracy: 98.5%** · **F1-Macro: 0.985**

Dataset: Synthetic (Kaggle) — 2,000 rows · 11 features · 3-class target

Business Problem

Kenya's grid is under mounting pressure from rapid e-mobility growth. The evening peak hit **2,288 MW** in October 2024 against an installed capacity of **3,199 MW** — leaving only **28% headroom**. E-mobility electricity consumption grew **300% YoY** (EPRA FY2024/25), with 6,442 registered EVs as of June 2025.

Unregulated charging during peak hours risks cascading failures. This model enables **demand-side management** by:

- Deferring **Low-priority** vehicles to off-peak hours (00:00–05:00), absorbing Kenya's 668.7 GWh/yr of curtailed geothermal surplus
- Ensuring **High-priority** vehicles (commercial fleets, emergency) charge immediately
- Estimated annual savings of **~KES 19.7M** for a 76-bus fleet via KPLC's E-Mobility TOU tariff (KES 7.9 off-peak vs KES 15.8 peak)

Limitation: The dataset is synthetic and may not capture real-world Kenyan grid noise (unplanned outages, substation-level variation). Real deployment should integrate live KPLC SCADA feeds and BasiGo fleet telemetry — see the Bonus section of the notebook for grounded scenario validation using real EPRA parameters.

Results Summary

Model	Accuracy	F1-Macro	Notes
Logistic Regression (baseline)	97.0%	0.971	Linear relationship captured well
Random Forest	87.7%	0.878	Underperformed baseline
XGBoost	94.3%	0.944	Strong but below LR
Decision Tree (tuned)	92.0%	0.921	GridSearchCV optimised
Calibrated LR (final)	98.5%	0.985	Isotonic calibration · Nested CV F1: 0.993

Logistic Regression outperforms tree-based models because the relationship between `initial_soc` , `station_load` , and charging priority is largely **linear and well-separated** — complex models added variance without improving accuracy.

Repository Structure

├─ EV_Charging_Optimization_Final.ipynb

├─ ev_charging_data.csv

├─ presentation.pdf

├─ final_logistic_pipeline_calibrated.pkl

├─ README.md

└─ .gitignore

Main analysis notebook (CRISP-DM)

Dataset – see Data Source below

Non-technical stakeholder slides

Deployment-ready model pipeline (joblib)

This file

Excludes .DS_Store, __pycache__, *.ipynb_checkpoints

Loading the saved model:

```
import joblib
model = joblib.load('final_logistic_pipeline_calibrated.pkl')
```

```
predictions = model.predict(X_new)      # Returns 0=Low, 1=Medium, 2=High
probabilities = model.predict_proba(X_new)
```

How to Run

1. Clone the repository

```
git clone <your-repo-url>
cd <repo-folder>
```



2. Install dependencies

```
pip install pandas numpy scikit-learn xgboost matplotlib seaborn joblib
```



3. Add the dataset — place `ev_charging_data.csv` in the root directory (see Data Source below)

4. Run the notebook

Open `EV_Charging_Optimization_Final.ipynb` → Kernel → Restart & Run All



Data Source

Dataset: EV Charging & Grid Optimization Data — Kaggle

(<https://www.kaggle.com/datasets/mjawad17/ev-charging-and-grid-optimization-data>) ←

Property	Value
Rows	2,000
Features	11
Target	charging_priority — 0=Low, 1=Medium, 2=High
Class balance	Low 33.8% · Medium 35.5% · High 30.8%

Key features used:

Feature	Role
initial_soc	Primary urgency signal — low SOC = higher priority
station_load	Grid stress — high load triggers deferral
electricity_price	Cost signal — charges align with off-peak windows
renewable_energy_ratio	Green charging opportunity
queue_length	Fairness / congestion signal
vehicle_type	Emergency and commercial vehicles get priority

Links

- Presentation: [presentation.pdf](#)
- Notebook: [EV_Charging_Optimization_Final.ipynb](#)

Real-world data sources (Bonus section):

- EPRA FY2024/25 Statistics Report: <https://www.epra.go.ke/statistics-0>

- KPLC Tariff Tracker: <https://www.stimatracker.com>